

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

*I D.Anand Paul (192425207) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.
3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.
5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are

then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

1. NETWORK DESIGN FOR A 20-PERSON STARTUP

1.1 Problem Statement

A startup with approximately 20 employees requires a reliable, simple, scalable, and easy-to-maintain computer network. The network should connect employee computers, printers, VoIP phones, laptops, wireless devices, and other office equipment.

For this requirement, star topology is the most suitable choice because every device is connected to a central network device such as a managed switch.

1.2 Selected Topology – Star Topology

In a star topology, every network device has an individual connection to a central switch. The central switch acts as the main communication point for the network.

A suitable network can use a 24-port Layer 2/Layer 3 managed switch. Although the startup has around 20 employees, additional ports are useful for printers, access points, VoIP phones, and future expansion.

1.3 Network Architecture

The basic architecture can be represented as:

Internet → Router/Firewall → Managed Switch → End Devices

The end devices may include:

- Employee computers
- Laptops
- Network printers
- VoIP phones
- Wireless access points
- Servers or network storage

Each wired device can be connected using Cat6 UTP Ethernet cable.

1.4 Main Components

Managed Switch

The managed switch is the central device of the star topology. It connects all computers and other network devices.

A 24-port switch provides enough ports for the current users and allows future expansion.

Router/Firewall

The router connects the internal office network to the Internet. It can perform:

- Internet connectivity
- DHCP
- Network Address Translation
- Routing
- Basic firewall functions

Wireless Access Point

A wireless access point can be connected to the switch to provide Wi-Fi connectivity to laptops, smartphones, and other wireless devices.

Cat6 UTP Cable

Cat6 UTP cables can provide Gigabit Ethernet connectivity for typical office distances. The maximum Ethernet channel length is generally designed around 100 metres.

1.5 Working

When an employee sends data to another computer, the data first reaches the central switch.

The switch examines the destination MAC address and forwards the frame to the appropriate port.

For Internet communication, the traffic is forwarded from the switch to the router/firewall and then to the Internet.

Therefore, the central switch manages communication efficiently between connected devices.

1.6 Fault Isolation

One major advantage of star topology is fault isolation.

For example, if one employee's Ethernet cable becomes damaged, only that employee's connection is affected.

The remaining employees can continue using the network.

This is much better than a topology where one backbone failure can affect the entire network.

1.7 Advantages

1. Easy to install and configure.
2. Easy to troubleshoot.
3. Failure of one cable affects only one device.
4. Easy to add new computers.
5. Centralised network management is possible.
6. Good performance with switched Ethernet.
7. Supports future expansion.
8. Suitable for modern office networks.

1.8 Limitations

1. Failure of the central switch can affect the connected devices.
2. More cable is required than in a traditional bus topology.
3. A good-quality switch increases initial cost.

1.9 Justification

Star topology is the best choice for a 20-person startup because it provides a balance between cost, reliability, performance, simplicity, and scalability.

The central switch makes network administration easier, while individual connections provide fault isolation.

2. HIGH-AVAILABILITY BANK NETWORK

2.1 Problem Statement

A bank requires a highly reliable network because banking applications involve financial transactions and sensitive customer information.

Network downtime can interrupt ATM services, online banking, branch operations, payment processing, and internal banking applications.

Therefore, the network must provide **high availability, redundancy, fault tolerance, and fast recovery**.

2.2 Selected Topology – Partial Mesh Topology

A **partial mesh topology** is highly suitable for the core network of a bank.

In a full mesh, every important device is directly connected to every other device. However, this can become expensive and complex as the number of devices increases.

Therefore, banks commonly use redundancy and multiple paths at critical network layers rather than connecting every device to every other device.

2.3 Basic Architecture

A simplified design is:

Internet/ISP 1 → Router 1 → Core Network

Internet/ISP 2 → Router 2 → Core Network

The core can contain multiple switches/firewalls with redundant connections.

Access and distribution devices can have redundant uplinks to two core switches.

2.4 Redundant Links

Redundant links provide alternative paths.

For example:

Access Switch → Core Switch 1

and

Access Switch → Core Switch 2

If the first link fails, traffic can use the second path.

This reduces the possibility of complete network failure.

2.5 Routing Protocols

Dynamic routing protocols such as **OSPF or BGP** can be used depending on the network layer and design.

OSPF can be used for internal routing, while BGP is commonly used for routing between autonomous systems and for Internet connectivity.

These protocols help the network select alternative paths when failures occur.

2.6 Fiber Optic Connectivity

Fiber optic cables are suitable for the bank's high-speed core network.

Two common types are:

- Single-mode fiber – suitable for longer distances.
- Multimode fiber – commonly used for shorter distances such as within data centers.

Fiber provides:

- High bandwidth
- Low latency
- Resistance to electromagnetic interference
- Long-distance communication capability

2.7 Additional Reliability Measures

Network redundancy alone is not enough for a bank.

Additional infrastructure can include:

- Dual power supplies
- UPS systems
- Backup generators

- Redundant firewalls
- Redundant routers
- RAID storage
- Backup servers
- Multiple Internet service providers
- Network monitoring systems

2.8 Failure Example

Suppose the primary router fails.

In a redundant architecture, the backup router can continue providing connectivity.

Similarly, if one core switch or link fails, traffic can be redirected through another available path.

This is called **fault tolerance**.

2.9 Advantages

1. Very high reliability.
2. Multiple communication paths.
3. Better fault tolerance.
4. Reduced downtime.
5. Suitable for critical financial services.
6. High-speed fiber connectivity.
7. Supports network redundancy.
8. Easier recovery from individual link failures.

2.10 Limitations

1. Higher installation cost.
2. More complex configuration.
3. Requires skilled network administrators.
4. More network hardware is required.
5. Maintenance and monitoring are more demanding.

2.11 Justification

A partial mesh is appropriate for a bank because banking services require continuous availability.

Redundant links, routers, switches, power systems, and Internet connections ensure that a single hardware or link failure does not necessarily stop critical services.

3. LOW-COST CLASSROOM NETWORK

3.1 Problem Statement

A small classroom computer laboratory may have a limited budget and a relatively small number of computers.

The network should allow students to share files and access common resources while keeping installation costs low.

For a traditional low-cost networking scenario, **bus topology** can be considered.

3.2 Selected Topology – Bus Topology

In a bus topology, all computers share a common communication backbone.

Each device is connected to the main cable, and data travels along this shared medium.

A traditional bus network uses coaxial cable.

3.3 Components

A classic coaxial bus network requires:

- Coaxial backbone cable
- BNC connectors
- T-connectors
- 50-ohm terminators
- Network interface cards
- Computers

The backbone cable runs through the classroom and individual computers connect to it.

3.4 Terminators

A terminator is placed at both ends of the bus cable.

The terminators absorb the electrical signal and help prevent signal reflections.

Without proper termination, network communication can become unreliable.

3.5 Working

When a computer transmits data, the signal travels along the common backbone.

All connected computers can detect the signal, but only the computer with the matching destination address processes the relevant data.

Because multiple computers share the same communication medium, collisions can occur when more than one computer transmits simultaneously.

3.6 Collision Problem

One important disadvantage of bus topology is the possibility of collisions.

If two computers transmit at the same time, their signals can interfere with each other.

Traditional Ethernet bus networks used **CSMA/CD** to manage shared-medium communication.

3.7 Failure Scenario

The major weakness of a bus topology is the backbone cable.

If the main cable is broken, a large portion or the entire network can become unavailable.

This makes bus topology less suitable for modern enterprise networks.

3.8 Advantages

1. Low hardware requirement.
2. Simple basic structure.
3. Less cable than star topology.
4. No central switch is required in the classic implementation.
5. Suitable for understanding shared-medium networking.
6. Lower initial cost for a very small traditional network.

3.9 Limitations

1. Backbone failure can affect the entire network.
2. Troubleshooting can be difficult.
3. Collisions can reduce performance.
4. Network performance decreases as traffic increases.
5. Adding or removing devices can affect the backbone.
6. Bus topology is largely obsolete for modern Ethernet LANs.

3.10 Important Technical Note

A modern Ethernet network using an unmanaged switch is **not technically a bus topology**. A switch with individual Ethernet cables creates a physical star topology.

Therefore, if the assessment specifically asks for bus topology, the traditional coaxial bus architecture with BNC connectors and terminators is the technically correct representation.

3.11 Justification

For a theoretical low-cost classroom scenario, bus topology demonstrates how multiple computers can share a common communication medium.

However, for an actual modern classroom network, an inexpensive Ethernet switch and Cat6/Cat5e cables would normally be preferred because it provides better reliability and performance.

4. UNIVERSITY CAMPUS NETWORK

4.1 Problem Statement

A university may contain thousands of users distributed across multiple buildings.

Examples include:

- Classrooms
- Laboratories
- Libraries
- Hostels
- Administrative offices
- Research centers
- Faculty offices
- Cafeterias

Because of the size and complexity of a university, one simple topology cannot efficiently serve the entire campus.

4.2 Selected Topology – Hybrid Topology

A **hybrid topology** combines two or more network topologies.

For a university campus, a suitable design can combine:

- Mesh or ring at the backbone
- Star topology inside buildings
- Hierarchical/tree structure for distribution and access

This provides both reliability and scalability.

4.3 Campus Backbone

The campus backbone can use high-speed fiber optic links between major network buildings.

A redundant ring or partial mesh can be used so that multiple paths are available.

For example:

Core 1 ↔ Core 2 ↔ Core 3 ↔ Core 4 ↔ Core 1

If one backbone link fails, traffic can potentially travel through another available path.

4.4 Three-Layer Hierarchical Model

A large campus network can be divided into:

Core Layer

The core layer provides high-speed backbone connectivity between major parts of the campus.

Distribution Layer

The distribution layer connects individual buildings or groups of access switches to the core.

Access Layer

The access layer connects end devices such as:

- PCs
- Printers
- IP phones
- Wireless access points
- IoT devices

4.5 Building-Level Design

Each building can have a distribution switch connected to the campus core using redundant fiber links.

Inside the building, access switches can connect to the distribution switch.

Individual computers can then connect to access switches using Cat6 Ethernet cables.

This creates a hierarchical star/tree structure.

4.6 Wireless Network

Wireless access points can be installed throughout the campus.

PoE-enabled Ethernet switches can supply both:

- Data connectivity
- Electrical power

to compatible access points through a single Ethernet cable.

This reduces the need for separate power connections near every access point.

4.7 Scalability

A major advantage of the hybrid design is scalability.

If the university constructs a new building, the new building can receive a distribution switch and fiber connection to the existing campus backbone.

The existing buildings do not need to be completely redesigned.

4.8 Advantages

1. Highly scalable.
2. Suitable for large campuses.

3. Provides backbone redundancy.
4. Easy to manage using hierarchical architecture.
5. Supports wired and wireless users.
6. High-speed fiber can connect buildings.
7. Individual building failures can be isolated.
8. New buildings can be added easily.

4.9 Limitations

1. Higher initial cost.
2. Requires professional network planning.
3. Fiber infrastructure can be expensive.
4. Network management is more complex.
5. Requires skilled administrators.

4.10 Justification

Hybrid topology is the best choice for a university because different parts of the campus have different requirements.

The backbone requires redundancy and high speed, while individual buildings require simple and manageable connectivity.

Therefore, combining mesh/ring, star, and hierarchical structures provides an effective campus-wide solution.

5. COMPANY NETWORK WITH MULTIPLE DEPARTMENTS

5.1 Problem Statement

A company with departments such as HR, Finance, Sales, and IT needs a network that allows communication while maintaining security and logical separation.

If all devices are placed in one large network, unnecessary broadcast traffic can increase and sensitive departmental information may be exposed.

Therefore, the network should use **VLANs, switches, routing, and proper IP addressing.**

5.2 Proposed Architecture

A hierarchical architecture can be used:

Internet

↓

Router/Firewall

↓

Layer 3 Core Switch

↓

Department Access Switches

↓

End Devices

Departments can have separate VLANs.

5.3 VLAN Design

Example:

Department	VLAN	Example Network
HR	VLAN 10	192.168.10.0/24
Finance	VLAN 20	192.168.20.0/24
Sales	VLAN 30	192.168.30.0/24
IT	VLAN 40	192.168.40.0/24

Each VLAN represents a logically separate broadcast domain.

5.4 Role of Layer 2 Switch

Layer 2 switches connect computers and other devices within a department.

For example, computers belonging to the HR department can be assigned to VLAN 10.

The switch forwards Ethernet frames based on MAC addresses.

5.5 Role of Layer 3 Switch

A Layer 3 switch can perform routing between different VLANs.

This process is called **inter-VLAN routing**.

For example:

HR VLAN → Layer 3 Switch → Finance VLAN

Communication between departments can therefore be controlled through routing and security policies.

5.6 Role of Router and Firewall

The router/firewall connects the internal company network to external networks such as the Internet.

The firewall can control traffic based on security rules.

For example, sensitive Finance systems can be restricted so that only authorised users can access them.

5.7 Security Benefits of VLANs

VLANs provide logical separation.

For example:

- HR devices can be placed in VLAN 10.
- Finance devices can be placed in VLAN 20.
- Sales devices can be placed in VLAN 30.
- IT devices can be placed in VLAN 40.

This reduces unnecessary broadcast traffic and provides a foundation for access-control policies.

5.8 IP Addressing and Subnetting

Each VLAN can use a separate IP subnet.

For example:

HR: 192.168.10.0/24

Finance: 192.168.20.0/24

Sales: 192.168.30.0/24

IT: 192.168.40.0/24

The default gateway for each VLAN can be configured on the Layer 3 switch.

5.9 Redundancy

For better reliability, redundant links can be added between important switches.

A company may also use:

- Backup Internet connection
- Redundant core switches
- Backup power
- Multiple firewall devices
- Link aggregation

These features reduce downtime.

5.10 Scalability

The architecture can easily support company growth.

If a new department is created, a new VLAN and subnet can be configured.

Additional switches and users can also be added without redesigning the entire network.

5.11 Advantages

1. Better departmental separation.
2. Improved security.
3. Reduced broadcast traffic.
4. Efficient network management.
5. Easy expansion.
6. Supports inter-VLAN routing.
7. Allows firewall-based access control.
8. Supports redundancy.
9. Provides structured IP addressing.
10. Suitable for medium and large organisations.

5.12 Limitations

1. Configuration is more complex than a basic LAN.
2. Requires managed switches.
3. Network administrators need VLAN and routing knowledge.
4. Incorrect configuration can cause connectivity problems.
5. Additional security policies are required.

5.13 Justification

A VLAN-based hierarchical architecture is suitable for a company with multiple departments because it provides **security, performance, manageability, and scalability**.

Each department can operate in a separate logical network while authorised communication between departments can be enabled through Layer 3 routing and firewall policies.

